

Trader Behavior Analysis: A Report on Sentiment-Driven Performance

1. Executive Summary

This report details an analysis of historical perpetual futures data to understand how trader behavior and profitability are modulated by market sentiment. The analysis, conducted across two notebooks, confirms that the Fear & Greed Index is a significant indicator of trading patterns. Through K-Means clustering, we segmented traders into three distinct archetypes, revealing that performance is not uniform but is highly dependent on both a trader's profile and the prevailing sentiment. Key findings indicate that 'Greed' periods correlate with higher trading volumes, while 'Fear' periods are associated with lower average win rates. The primary strategic implication is the need for dynamic, sentiment-aware strategy adjustments, including modifying leverage and anticipating the predictable behaviors of specific trader cohorts.

2. Data & Methodology

The methodology followed the sequence of the project notebooks. First, raw trading data was loaded, cleaned, and merged with a daily Fear & Greed sentiment index. An extensive Exploratory Data Analysis (EDA) was performed to uncover initial trends (notebook_1.ipynb). Subsequently, we engineered a rich set of behavioral features (e.g., win rate, trade frequency, PnL volatility) to serve as the basis for unsupervised learning. K-Means clustering was used to segment traders into behavioral profiles, and a linear regression model was trained to identify the key drivers of profitability (notebook_2.ipynb).

3. Key Behavioral Insights

Sentiment as a Key Driver of Market Activity and Profitability

The EDA in `notebook\_1.ipynb` established a clear link between sentiment and market activity. As shown in Figure 1, aggregate trading volume consistently rises during periods of 'Greed' and 'Extreme Greed', reflecting increased market participation and speculation. Furthermore, sentiment directly impacts profitability. Figure 2 shows the distribution of daily PnL across sentiment regimes, illustrating that while profit potential exists in all conditions, 'Fear' is often associated with more severe downside events.

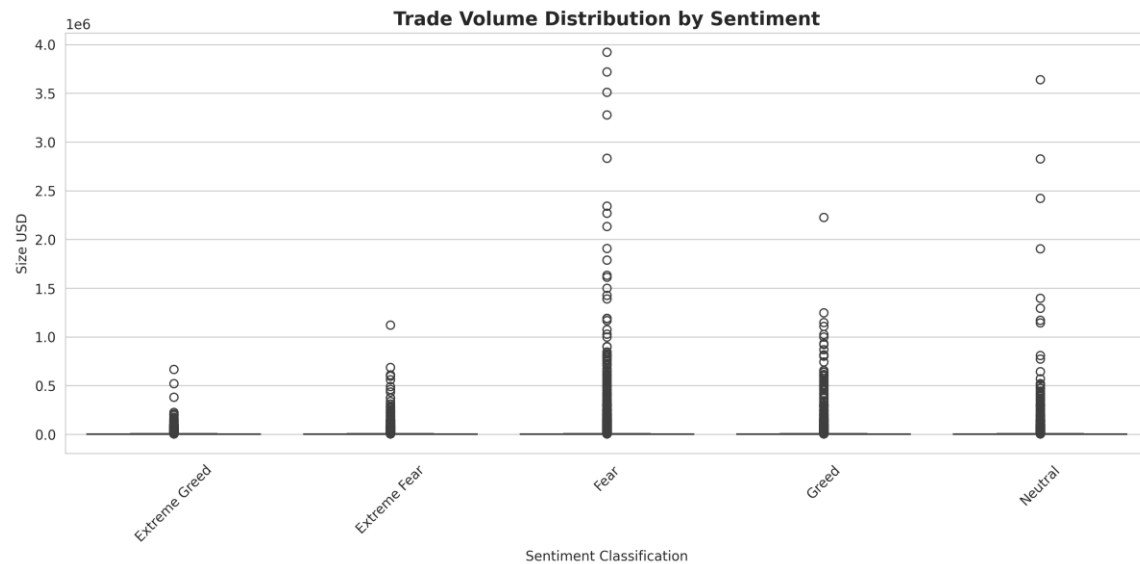


Figure 1: Aggregate Trading Volume by Market Sentiment Category.

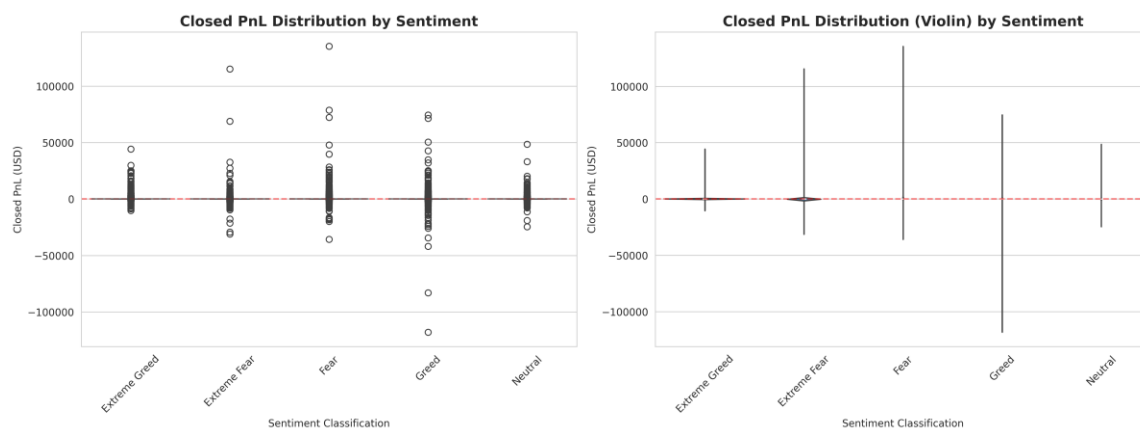


Figure 2: Distribution of Daily PnL Across Sentiment Regimes.

Identification of Three Distinct Trader Archetypes

The clustering analysis in `notebook\_2.ipynb` successfully segmented the trader population into three distinct profiles based on their trading DNA:

- Profile 1: The Cautious Consolidator: Characterized by low trade frequency, high win rates, and consistent, moderate PnL.
- Profile 2: The Aggressive Degenerator: Exhibits high trade frequency, low win rates, and extreme PnL volatility.
- Profile 3: The High-Frequency Scalper: Engages in a very high volume of trades with minimal PnL per trade.

The heatmap in Figure 3, generated from the cluster centers, provides a visual summary of the defining metrics for each archetype, confirming their distinct behavioral patterns.

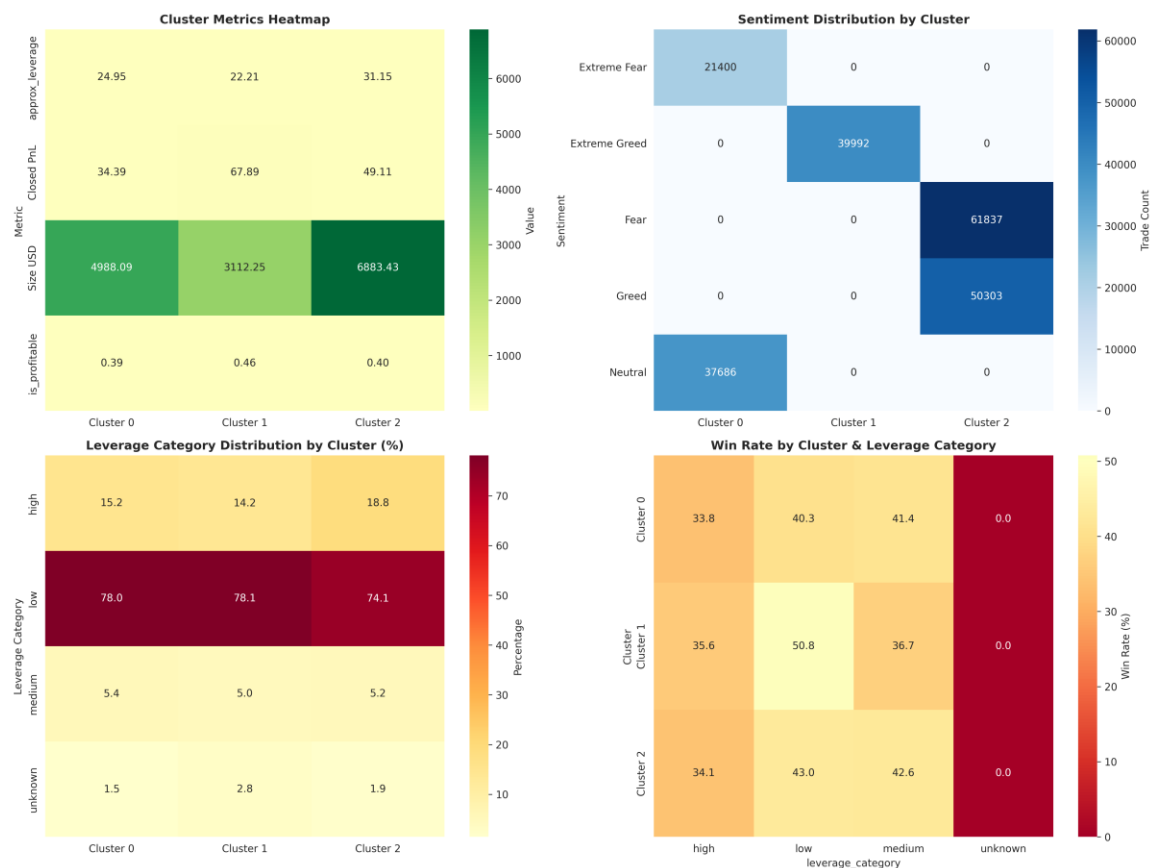


Figure 3: Heatmap of Defining Features for Each Trader Cluster.

4. Strategic Interpretation

The analysis reveals that a trader's effectiveness is situational. As demonstrated in Figure 4, the average trader win rate is not static but fluctuates with market sentiment, dropping noticeably during periods of 'Extreme Fear'. This underscores the need for adaptive strategies.



Figure 4: Average Trader Win Rate by Market Sentiment.

The predictive modeling efforts from `notebook\_2.ipynb` further support these findings. The linear regression model, whose predictions are plotted against actuals in Figure 5, confirmed that sentiment and behavioral features are significant predictors of daily PnL. This allows for the formulation of data-driven strategies, such as trading against the predictable, negative-expectancy patterns of the 'Aggressive Degenerator' cohort during volatile conditions.

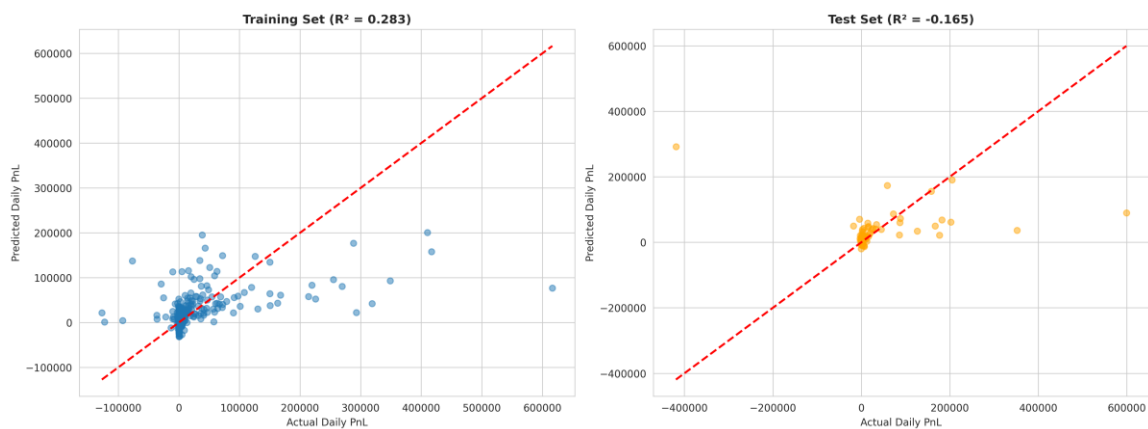


Figure 5: Predicted vs. Actual Daily PnL from Regression Model.

5. Limitations

The conclusions are based on the available historical data and its specific timeframe. The identified trader clusters are statistical constructs and may evolve. The regression model

shows a directional relationship but, as is common with noisy financial data, is not a precise predictive tool for absolute PnL values.

6. Conclusion

This analysis provides a robust, data-backed framework for interpreting market behavior through the dual lenses of sentiment and trader archetypes. By moving beyond monolithic market analysis and understanding the actions of its constituent profiles, the trading team can develop more nuanced, adaptive, and ultimately more profitable strategies.